AgroBot

Capstone-I Project Report

IoT-Based Soil Moisture Detection and Monitoring System

Submitted by

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Declaration

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Abstract

This report presents AgroBot, an IoT-based soil moisture detection and monitoring system developed as a Capstone-I project. The system addresses the widespread problem of poor irrigation management in Indian agriculture, where both over-irrigation and under-irrigation lead to significant crop losses. The device consists of a soil moisture sensor, an Arduino Uno microcontroller, an ESP8266 Wi-Fi module, and a piezoelectric buzzer, all assembled into a single compact unit. The microcontroller reads analog moisture values, converts them to a percentage, triggers local buzzer alerts when moisture goes outside safe thresholds, and transmits data wirelessly to a web server. A web dashboard displays live readings and a seven-day trend chart. This report covers the project objectives, components, system design, and planned future extensions.

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Chapter 1: Introduction

Water is vital for plant growth, yet it is often poorly managed in Indian agriculture. Both over-irrigation and under-irrigation lead to significant crop losses across regions like Maharashtra and Madhya Pradesh. Farmers frequently overwater during monsoons, causing root diseases, or under-irrigate in summers, leading to crop failure. Widespread yield losses due to improper watering practices have been reported extensively. A key issue is the lack of affordable, real-time soil moisture monitoring, forcing farmers to rely on guesswork, which often results in inefficient irrigation and preventable damage.

AgroBot addresses this problem using IoT (Internet of Things). The idea is that physical devices like sensors can be connected to the internet and share data automatically without any manual effort. The device reads the moisture level of the soil using a moisture sensor and sends this data to a website through a Wi-Fi module. The user can then open the website from any device and see the current moisture level without having to go near the field.

The microcontroller used in this project is the Arduino Uno. It is built around the ATmega328P chip and acts as the brain of the system — it reads the sensor data, processes it, checks if it is within safe limits, triggers the buzzer if not, and tells the Wi-Fi module to send the data to the internet.

All components — the Arduino Uno microcontroller, the soil moisture sensor, the ESP8266 Wi-Fi module, and the buzzer — are assembled together as one single compact device. The website displays a live reading of the current soil moisture level and a weekly report showing how the moisture has changed over the past seven days.

Chapter 2: Objectives and Scope

Objectives

The main objective of this project is to build a low-cost, compact, and functional IoT-based soil moisture detection and monitoring system that provides real-time data to the user without requiring physical presence at the field. The specific objectives are:

1. Design and assemble a single compact hardware device consisting of the Arduino Uno microcontroller, a resistive soil moisture sensor, an ESP8266 Wi-Fi module, and a piezoelectric buzzer.
2. Program the Arduino Uno to read analog moisture values via the on-chip ADC, convert them to a moisture percentage, and trigger the buzzer automatically when moisture crosses defined thresholds.
3. Establish wireless data transmission from the Arduino to a remote web server using the ESP8266 Wi-Fi module and HTTP POST requests.
4. Build a website that displays the live soil moisture reading and generates a weekly report showing moisture trends over the past seven days.
5. Provide an immediate local alert at the device level using the buzzer, so that even without internet access, the user is warned when the soil condition is critical.

Scope

The scope covers the complete design, development, and testing of a prototype soil moisture monitoring system for agricultural use. This includes hardware circuit assembly, Arduino microcontroller programming with buzzer alert logic, physical device enclosure, and full website integration with server-side data storage.

The project is designed and tested as a single-node system, meaning one device monitors one location. Multi-node deployment and large-scale farm coverage are outside the scope of this work. The farm security system using ultrasonic sensors and the automated drainage system are identified as future extensions.

Chapter 3: Components & Technical Description

This chapter describes each component used in the project, what it does technically, and how it connects to the rest of the system.

Arduino Uno Microcontroller

The Arduino Uno is the central microcontroller of the device, built around the ATmega328P chip. It has 14 digital GPIO pins and 6 analog input pins. The analog pin A0 is connected to the output of the moisture sensor. The on-chip ADC converts the continuous analog voltage from the sensor into a 10-bit digital number between 0 and 1023. The program maps this number to a moisture percentage and communicates with the ESP8266 Wi-Fi module. The board is powered via USB or a 5V power adapter.

Soil Moisture Sensor

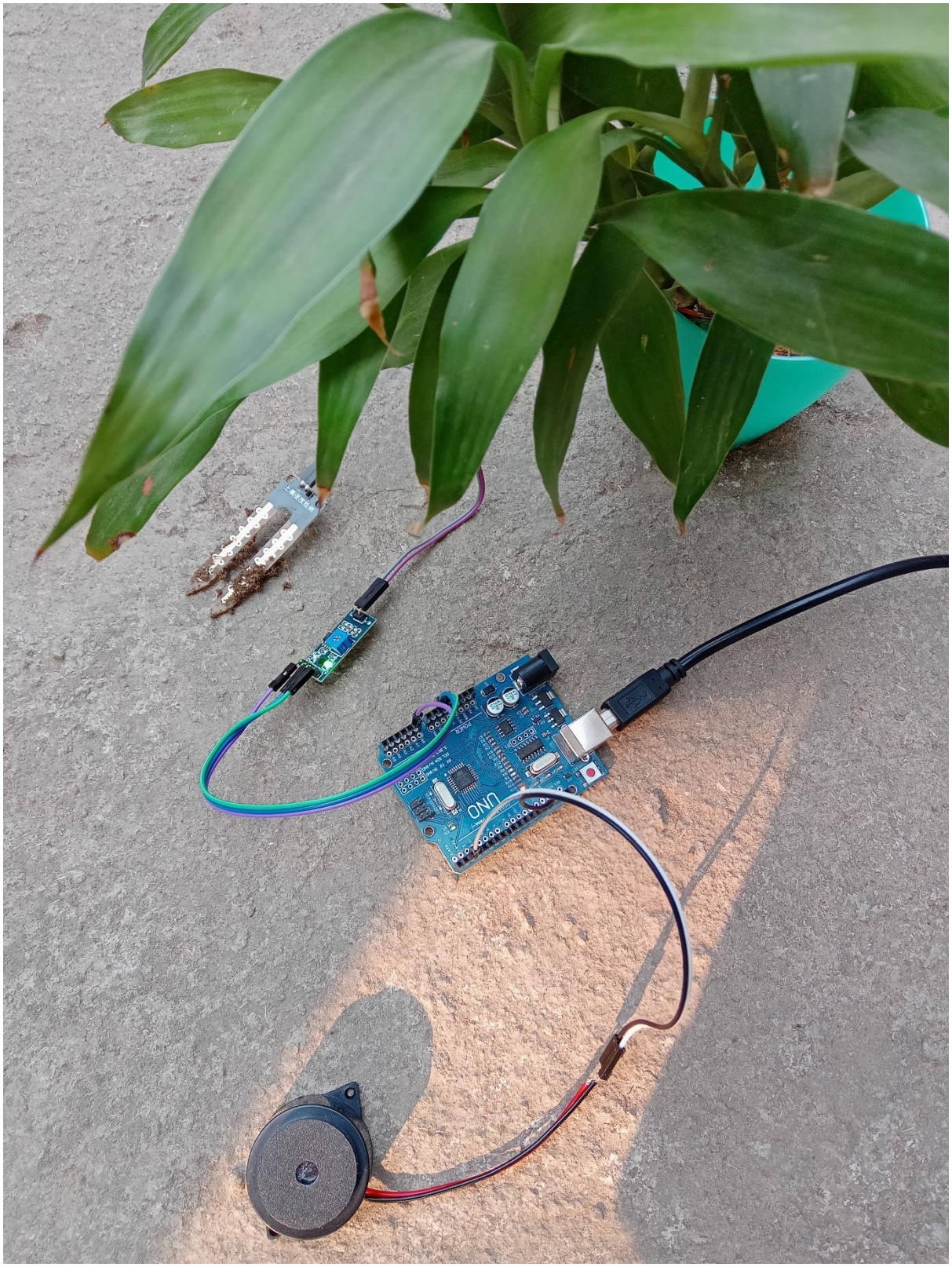
The soil moisture sensor is a resistive type sensor with two metal probes and a small control board. The probes are pushed into the soil and measure how easily current can flow between them. Wet soil conducts electricity better (lower resistance), while dry soil has higher resistance. The control board converts this into an analog output voltage between 0V and 5V, which is connected to pin A0 on the Arduino.

ESP8266 Wi-Fi Module

The ESP8266 is a Wi-Fi transceiver module with a built-in TCP/IP stack, used purely as a wireless communication module. It connects to the Arduino Uno via UART serial pins and receives instructions through AT commands sent by the Arduino. These commands tell it to connect to a Wi-Fi network and send an HTTP POST request to the web server carrying the moisture reading.

Buzzer (Piezoelectric)

A piezoelectric buzzer provides an immediate local alert when the moisture level is out of the safe range. It is connected to a digital GPIO output pin on the Arduino. Two threshold values are defined in the code — a lower limit and an upper limit. If the reading falls below the lower limit or rises above the upper limit, the Arduino sends a HIGH signal to the buzzer pin. When moisture returns to a safe level, the Arduino sends a LOW signal and the beeping stops. This alert works even without Wi-Fi connectivity.


UNO

Chapter 4: System Design & Methodology

Hardware Design

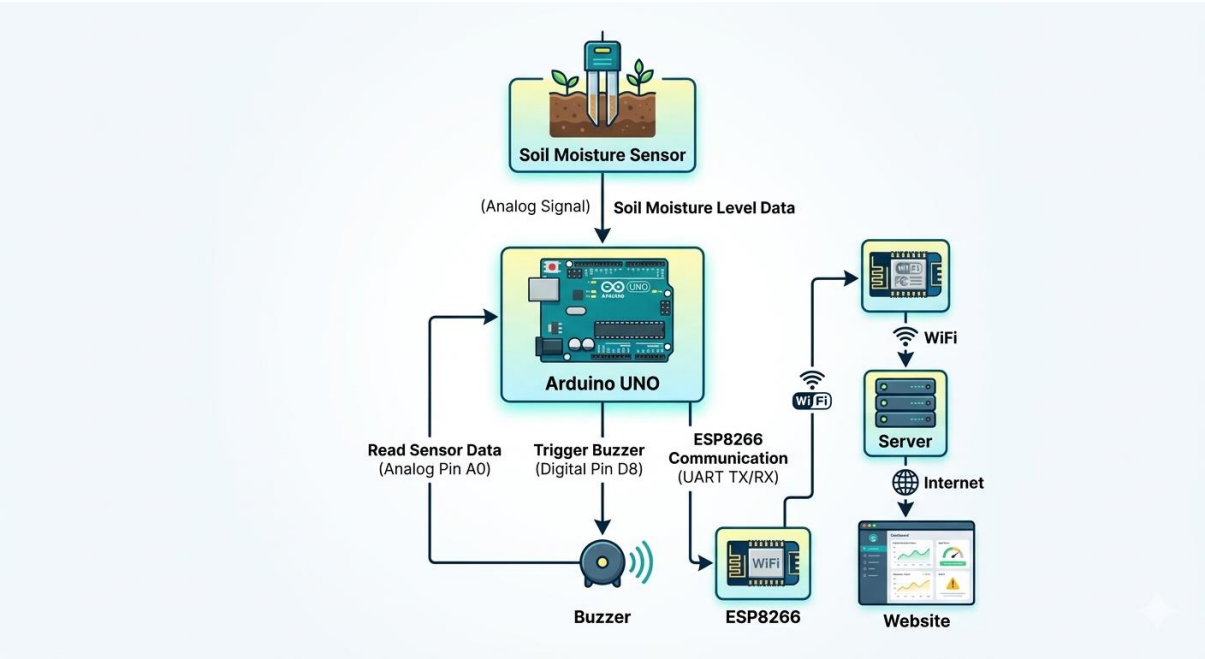
The soil moisture sensor is pushed into the soil and its analog output is connected to pin A0 on the Arduino Uno. The buzzer is connected to digital output pin 8. The ESP8266 Wi-Fi module is connected to the Arduino using the TX and RX UART serial pins. The entire circuit runs on 5V and can be powered by a USB connection. All components are mounted together in a single enclosure.

When powered on, the Arduino continuously runs the program in a loop. Every few seconds it reads the analog value from the moisture sensor on pin A0. The on-chip ADC converts this voltage to a number between 0 and 1023, which is mapped to a moisture percentage. The program checks this percentage against defined thresholds — if outside the safe range, pin 8 is set HIGH, activating the buzzer. Once moisture returns within range, pin 8 is set LOW.

After the threshold check, the Arduino sends AT commands to the ESP8266 module over UART. These commands instruct the ESP8266 to connect to the Wi-Fi network and send the current moisture reading to the web server via HTTP POST.

Data Flow

6. Soil Moisture Sensor detects water content and outputs an analog voltage.
7. Arduino Uno reads the analog signal via ADC, converts it to a moisture percentage.
8. ESP8266 Wi-Fi Module receives data via UART and transmits it to the web server using HTTP POST.
9. Web Server receives and stores the data in a database.
10. Website Dashboard pulls and displays live readings and weekly trend charts.


Soil Moisture Sensor
(Analog Signal)
Soil Moisture Level Data
Arduino UNO
Read Sensor Data
(Analog Pin A0)
Trigger Buzzer
(Digital Pin D8)
ESP8266
Communication
(UART TX/RX)
WiFi
WiFi
Server
Internet
Buzzer
ESP8266
Website

Chapter 5: Conclusion & Future Work

Conclusion

The circuit connecting the Arduino Uno microcontroller, the soil moisture sensor, the ESP8266 Wi-Fi module, and the piezoelectric buzzer has been successfully assembled. The core Arduino program correctly reads the analog moisture value through the ADC, converts it to a percentage, and triggers the buzzer when moisture goes above or below the defined thresholds. The basic front-end layout of the website has been drafted. The remaining work — Wi-Fi data transmission, server-side data storage, and full website integration — is planned for completion. Overall, the project is progressing well and the complete system is expected to be delivered on schedule.

Future Work

The immediate next steps are to complete Wi-Fi data transmission, set up server-side data storage, and integrate the full website with live readings and the weekly report. Beyond this, the following extensions are planned:

- Automated Drainage System — A relay-controlled drainage valve activated by the microcontroller when moisture exceeds the upper threshold, preventing soil waterlogging without manual intervention.
- Farm Security System — HC-SR04 ultrasonic sensors placed along the farm boundary to detect intrusions by animals or persons.
- Animal Deterrent Devices — On intrusion detection: piezoelectric alarm, PIR-activated floodlight, relay-controlled electric fence pulse, SIM800L GSM SMS alerts, and a servo-controlled water sprinkler.
- Multi-Node Deployment — Support for multiple sensor nodes covering larger farm areas.
- Field Testing — Real-world deployment and performance testing under actual agricultural conditions.

References

[1] Arduino Official Documentation. https://www.arduino.cc

[2] Soil Moisture Sensor Module — Working and Pinout. Components101, 2021.

[3] ESP8266 AT Instruction Set and Wi-Fi Module Guide. Espressif Systems. https://www.espressif.com

[4] IoT Based Soil Moisture Monitoring System for Smart Irrigation. IJERT, Vol. 9, 2020.

[5] Sensor Partners — IoT Sensor Resources. https://sensorpartners.com

[6] Dainik Bhaskar (2021) — Yield loss due to incorrect watering practices among small and marginal farmers in Vidarbha.

[7] The Times of India (2022) — Crop loss across Maharashtra and Madhya Pradesh due to improper irrigation.

[8] Lokmat (2023) — Onion and tomato crop losses in Nashik district due to waterlogging from unmonitored drip irrigation.